ASIRIVE // CORTEX

TKKYIA Interview //
Hybrid Edge-Server Wearable

ASIRIVE » CORTEX

Navigation
INDEPENDENCE

Made by Haziq, Irfan, Eryna

Designed with SAVH Singapore

https://www.asirive.com


THE WHITE CANE
COVERS MOST HAZARDS
BUT
SOME
HAZARDS CAN’T BE DETECTED BY
THE CANE

FOR EXAMPLE

Cane – Covers

Ground – Curbs,

Steps, Holes

Cane – Invisible

Overhang · Stairs

going up ·

Approaching Vehicles

ASIRIVE // CORTEX

OTHER OPTIONS DONT WORK EITHER

Phone Trap

App need Hands. Cane goes in the pocket

Price Lock

Premuim AI glasses cost more than most can pay

Too Few Tools

Many Need Help. Few cover the right Hazards

ASIRIVE // CORTEX

WE TRIED TO REPLACE THE CANE

SAVH GAVE US FEEDBACK



VERSION 1

Full 3D Spatial Audio → Replace the Entire Cane

THE PIVOT

We went to SAVH for REAL Answers → Overwhelming. Distracting. We need to hear surroundings

VERSION 2

Cane-aware safety — AI fills the cane-invisible gaps, never duplicates what the cane does

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WE TRIED TO REPLACE THE CANE

SAVH GAVE US FEEDBACK



V.2

Cane-aware safety — AI fills the cane-invisible gaps, never duplicates what the cane does

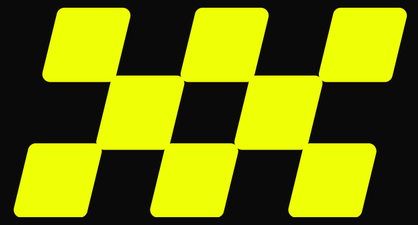
HAZARD	CANE?	ASIRIVE CORTEX?
GROUND	✓	✗
OVERHEAD	✗	✓
STAIRS UP	✗	✓
OBJECS	✗	✓

FEWER FALSE ALERTS. FASTER RESPONSE. LOWER COGNITIVE LOAD.

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5 SPECIALISTS

1 RULE : SAFETY NEVER LEAVES THE DEVICE



ACTION VOICE

PERCEPTION EYES

Cane–Aware
Safety Fusion

<100ms
Local
Always on

CONTEXT MEMORY

Remembers
your places and
recent
conversations

Passive
Local

DECISION JUDGMENT

Powered by Gemini 3.1 Flash Live

Audio IN. Audio Out

<500ms –Cloud
On Demand

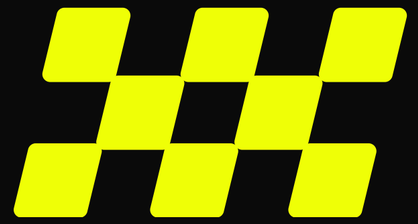


https://www.asirive.com

ASIRIVE // CORTEX

5 SPECIALISTS

1 RULE : SAFETY NEVER LEAVES THE DEVICE



ACTION VOICE

Speak Nav + Alerts

Two Voices

Chat & Safety Voice

Always Distinct

FEEDBACK LOOP

Logs every Detection

Learns over time

Async

Local & Cloud

https://www.asirive.com

COMPONENT	SPEC
• COMPUTE	• RPI5 + Hailo
• CAMERA	• Camera Module 3
• AUDIO	• BT Glasses (mic +speaker
• POWER	• 20000mah (~1 day)
• WEIGHT	• ~300g

$250 WEARABLE. 1/22TH OF THE COST OF ORCAM

COMMODITY HARDWARE »

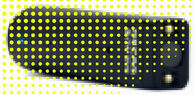
Three Ways Cortex Beats OrCam

Build for Navigation

- Turn by Turn – LTA buses – Saved Places

Cane Aware Safety

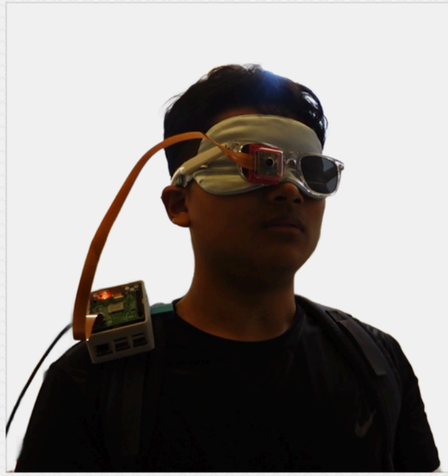
Open Architecture



OrCam MyEye 3 Pro

~~$4490~~ $4250

VS



Asirive Cortex

$250

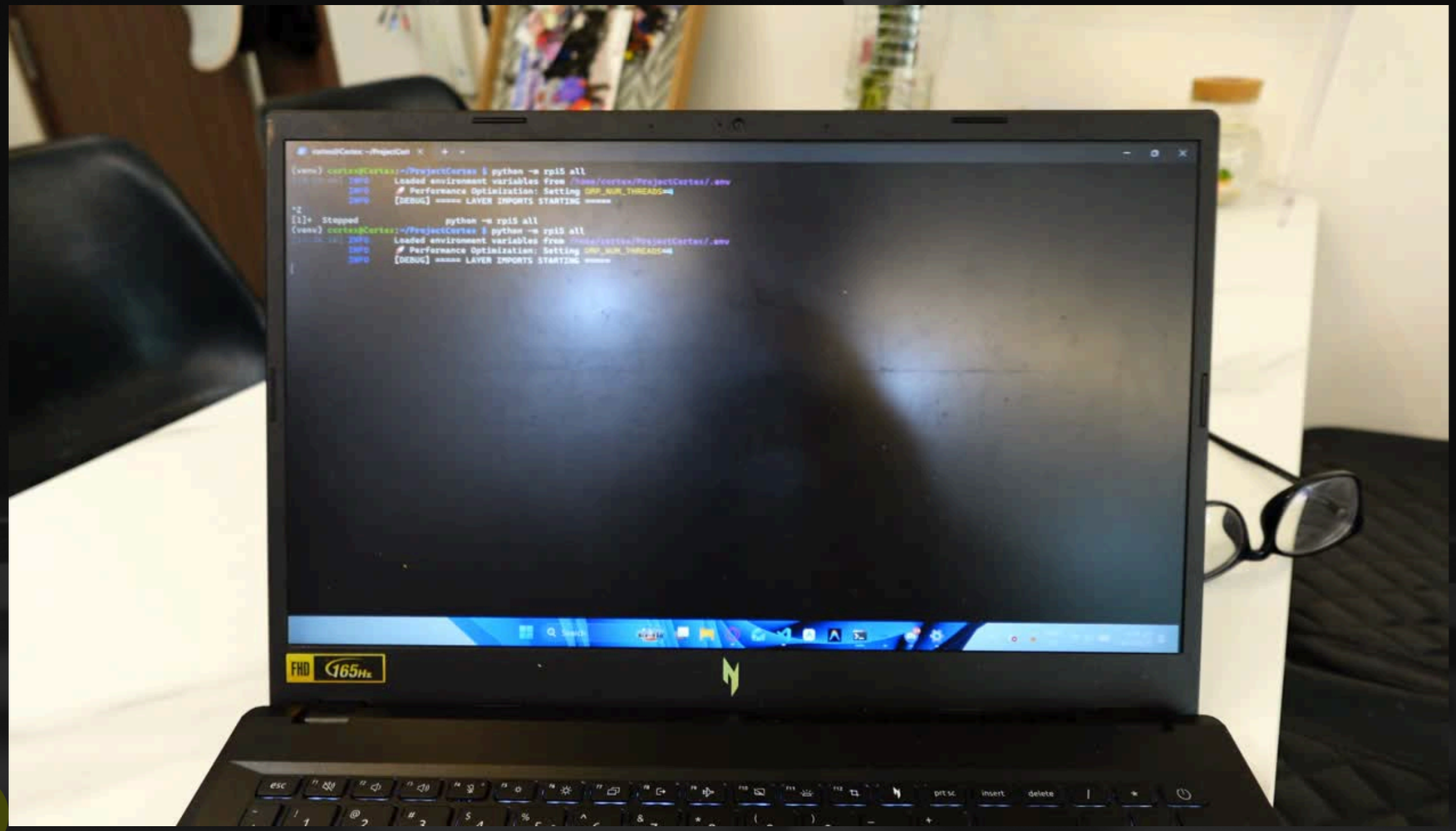


LIVE DEMO

TUI + PROTOTYPE

LIVE DEMO

TUI + PROTOTYPE – VIDEO (SUBMITTED)



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VERSION 1

Monocular Cam

SAVH Validated

Cane Aware Locked

$250



VERSION 2

Binocular Camera + TOF Sensor

More Tests with SAVH

Better Ergonomics

Development of Full Local Navi



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VERSION 3

SAVH Stress Test

Asitive Ecosystem – App

Institutional Pilot

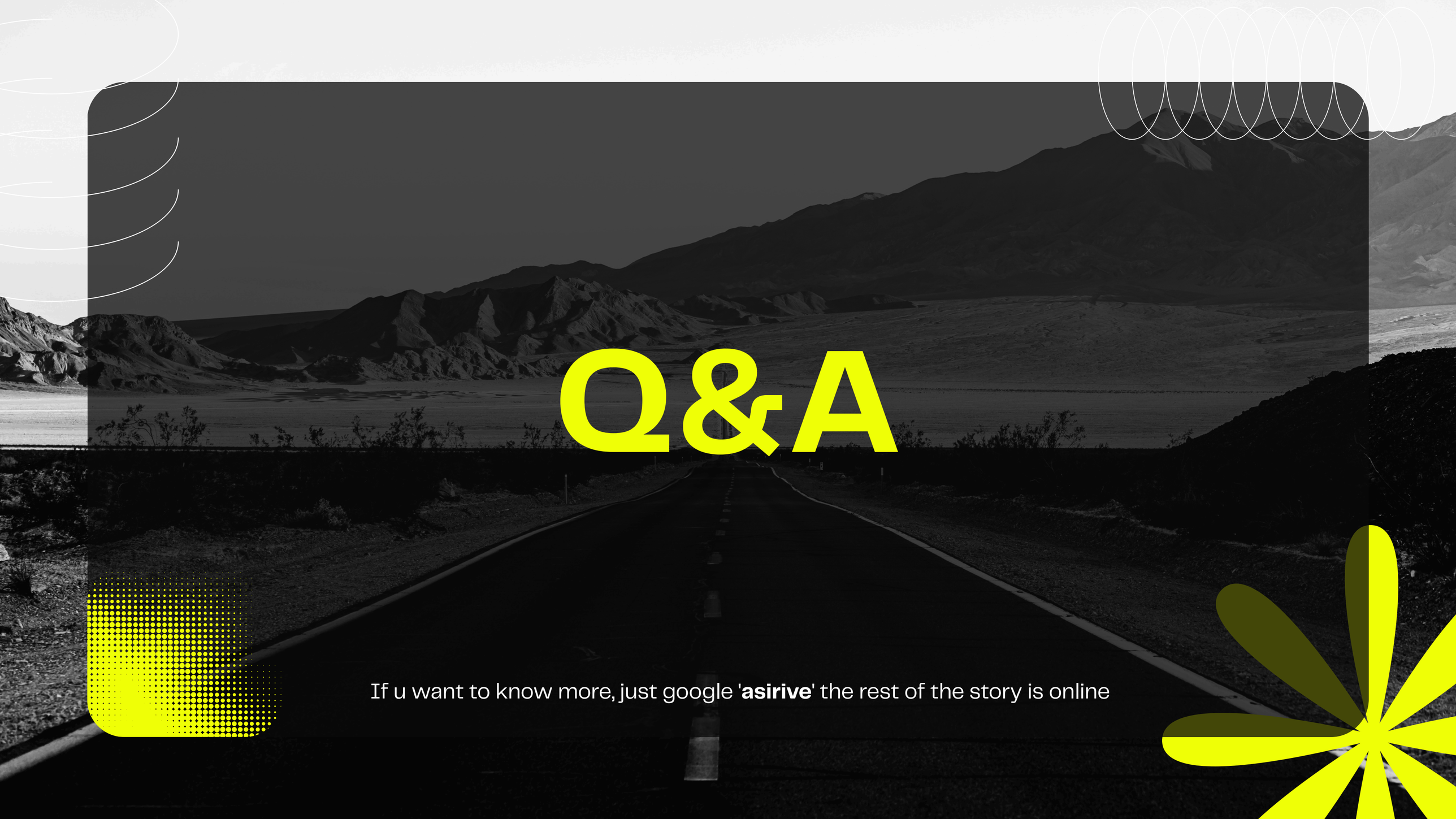
More Units for Testing

Release of Full Local Navi

Improved Ergonomics

YIA WAS A CHECKPOINT

BUT, THE BUILD CONTINUE



Q&A

If u want to know more, just google '**asirive**' the rest of the story is online

Breaking the expensive monopoly with hybrid edge-server architecture

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THANK YOU! »»



